



MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 柔軟劑
ECO-STAR CLEARLY SOFT

Other means of identification : Not applicable.

Recommended use : Laundry softener

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : Product is sold ready to use.

Company : Ecolab Limited
15/F, Lu Plaza, 2 Wing Yip Street
Kwun Tong, Kowloon Hong Kong
+852-2372 7768

Emergency telephone number : +852-2372 7768

Issuing date : 07.03.2017

Section: 2. HAZARDS IDENTIFICATION

Hazard classification : Xi - IRRITANT

R-phrases(s) : R36 - Irritating to eyes.
R52/53 - Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

S-phrases(s) : S26 - In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S60 - This material and/or its container must be disposed of as hazardous waste.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration: (%)
Amines (incl. ethanolamines)	68410-69-5	10 - 30
propan-2-ol	67-63-0	1 - 5
Silicones	68554-54-1	1 - 5

Section: 4. FIRST AID MEASURES

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth with water.

If inhaled : Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Section: 5. FIREFIGHTING MEASURES

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : None known.

Specific hazards during firefighting : Not flammable or combustible.

Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
nitrogen oxides (NOx)

Special protective equipment for firefighters : Use personal protective equipment.

Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. Do not flush into surface water or sanitary sewer system.

Section: 7. HANDLING AND STORAGE

Advice on safe handling : Avoid contact with skin and eyes. Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 10 °C to 40 °C

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
propan-2-ol	67-63-0	OEL-TWA	400 ppm 983 mg/m ³	HK OEL
		OEL-STEL	500 ppm 1,230 mg/m ³	HK OEL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety glasses with side-shields

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: liquid
Colour	: opaque, off-white
Odour	: Perfumes, fragrances
pH	: 6.5 - 7.5, 100 %
Flash point	: 117 °C closed cup, Does not sustain combustion.
Odour Threshold	: no data available
Melting point/freezing point	: no data available
Initial boiling point and boiling range	: > 100 °C
Evaporation rate	: no data available
Flammability (solid, gas)	: no data available
Upper explosion limit	: no data available
Lower explosion limit	: no data available
Vapour pressure	: no data available
Relative vapour density	: no data available
Relative density	: 0.98 - 1.04
Water solubility	: insoluble
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available

MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Viscosity, kinematic	: no data available
Explosive properties	: no data available
Oxidizing properties	: no data available
Molecular weight	: no data available
VOC	: no data available

Section: 10. STABILITY AND REACTIVITY

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: Bases
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NOx)

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes	: Causes serious eye irritation.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact	: Redness, Pain, Irritation
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product

Acute oral toxicity	: no data available
Acute inhalation toxicity	: no data available
Acute dermal toxicity	: no data available

MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Skin corrosion/irritation : No skin irritation
Serious eye damage/eye irritation : No eye irritation
Respiratory or skin sensitization : no data available
Carcinogenicity : no data available
Reproductive effects : no data available
Germ cell mutagenicity : no data available
Teratogenicity : no data available
STOT - single exposure : no data available
STOT - repeated exposure : no data available
Aspiration toxicity : no data available

Components

Acute oral toxicity : Amines (incl. ethanolamines)
LD50 rat: > 15,000 mg/kg

propan-2-ol
LD50 rat: 5,840 mg/kg

Silicones
LD50 rat: > 5,000 mg/kg

Components

Acute inhalation toxicity : propan-2-ol
4 h LC50 rat: 30 mg/l

Silicones
4 h LC50 rat: 978 mg/l

Components

Acute dermal toxicity : propan-2-ol
LD50 rabbit: 12,870 mg/kg

Silicones
LD50 rabbit: > 2,000 mg/kg

Section: 12. ECOLOGICAL INFORMATION

Ecotoxicity

Product

Toxicity to fish : 96 h LC50: 7.14 mg/l
Toxicity to daphnia and other aquatic invertebrates : no data available
Toxicity to algae : no data available

Components

Toxicity to daphnia and other aquatic invertebrates : propan-2-ol
LC50 Daphnia magna (Water flea): > 10,000 mg/l

Persistence and degradability

MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Poorly biodegradable

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

The product should not be allowed to enter drains, water courses or the soil.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (CN_DG)

Not dangerous goods

Sea transport (IMDG/IMO)

Not dangerous goods

Section: 15. REGULATORY INFORMATION

National Regulations

Authored with regard to the Factories & Industrial Undertakings (Dangerous Substances) Regulations.

The components of this product are reported in the following inventories:

Switzerland. New notified substances and declared preparations :

not determined

United States TSCA Inventory :

On TSCA Inventory

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL.

Australia. Industrial Chemical (Notification and Assessment) Act :

not determined

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :

not determined

MATERIAL SAFETY DATA SHEET

ECO-STAR CLEARLY SOFT

Japan. ENCS - Existing and New Chemical Substances Inventory :

On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :

not determined

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

On the inventory, or in compliance with the inventory

China Inventory of Existing Chemical Substances :

On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :

On the inventory, or in compliance with the inventory

Section: 16. OTHER INFORMATION

Issuing date	: 07.03.2017
Date of first issue	: 07.03.2017
Version	: 1.0
Prepared by	: Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.